AWS FOR ENTERPRISE APPS

Running Microsoft workloads on AWS: Your questions answered

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Your top 10 questions—answered

Amazon Web Services (AWS) has been helping customers migrate their Microsoft workloads to the cloud since 2008—longer than any other major cloud provider. We've earned the trust of millions of active customers who are now experiencing better performance and reliability, greater security, a lower total cost of ownership (TCO), and flexible licensing options for their Windows Server, SQL Server, and .NET workloads.

From lifting and shifting workloads to moving entire data centers, AWS provides the organizational, operational, and technical capabilities you need for a successful migration to the cloud. With the deepest set of migration and modernization services, programs, and partners that specialize in Microsoft applications, you can start realizing the business value of AWS quickly.

In this eBook, we answer 10 questions we're often asked about running Microsoft workloads on AWS—ranging from why AWS to the licensing, migration, and optimization options for your Windows Server, SQL Server, and .NET workloads. We've also included our top recommendation for any customer running these workloads on AWS.

Let's get started.

1

Why should we choose AWS to run our Microsoft workloads in the cloud?

When experience matters, customers choose AWS.

We have been investing in making AWS the best place to run Microsoft workloads since 2008—longer than any other major cloud provider—so that we can provide you with the most operational experience, at greater scale, of any cloud.

When license flexibility matters, customers choose AWS. Let's face it—Microsoft licenses are expensive. This is where we can help. AWS offers flexible licensing options for your Microsoft workloads, allowing you to bring your own licenses or pay as you go without signing any contracts or agreements.

And when modernization matters, customers choose AWS. Replace your Microsoft-licensed products and applications with specialized, built-for-the-cloud technologies and break free from the punitive terms and high cost of Microsoft licensing.

We've helped thousands of customers migrate and modernize their Microsoft workloads with AWS. We can help you, too.

2

Can we bring our Microsoft licenses to AWS?

Yes, you can bring your own licenses. It's called BYOL.

AWS is a Microsoft Authorized License Mobility Partner, so if you have licenses with License Mobility—such as SQL Server with active Software Assurance—you can bring them to Amazon Elastic Compute Cloud (Amazon EC2) on shared tenancy.

BYOL if you want to:

- Take advantage of cloud efficiencies and leverage your existing licensing investments
- Extend the lifecycle of your software without additional hardware costs
- Expedite your migration to the cloud by using existing virtual machine images

You can also use AWS instances that include the cost of licensing.

By using AWS-provided licenses on Amazon EC2 or Amazon Relational Database Service (Amazon RDS) instances, your Windows Server, SQL Server, Office, and Visual Studio licenses will always be fully compliant.

Choose AWS-provided licenses if you want to:

- Pay as you go with no upfront costs or long-term commitments
- Deploy workloads that scale up and down as needed
- Fully control when you are being billed for license-included instances
- Get out of the business of managing licensing compliance—AWS does it for you

Significantly reduce licensing costs through cloud optimization whether you BYOL or use AWS-provided licenses. **The more your servers/instances are rightsized for the cloud, the less you pay.**

3

How can we reduce our dependency on Microsoft licensing with AWS?

We get this question all the time. New and existing customers alike are looking for ways to break free from the punitive terms and high cost of Microsoft licensing.

Customers want to lower their costs, harden their security, and improve the price performance of their Microsoft applications. This is where AWS can help. With AWS, you can speed up innovation by replacing your Microsoft-licensed products and applications with open-source alternatives and technology built specifically for the cloud.

Here are some ways our customers are reducing their licensing dependencies:

- Moving from Windows Server to Linux
- Porting applications from .NET Framework to .NET
- Decomposing monoliths into microservices
- Implementing DevOps with container and serverless technologies
- Transitioning your data tier to Amazon Aurora and other purpose-built databases

If you're ready to move off legacy licensing and experience ultimate freedom and savings, AWS is ready to assist in your modernization journey.

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How can we optimize the cost of our Windows Server workloads on AWS?

AWS offers more than 750 generally available secure and resizable compute instances—more than any other major cloud provider—with a wide selection of instance types optimized to fit different use cases, including CPU, memory, storage, and networking capacity.

While this gives you the flexibility to choose the appropriate mix of resources for your applications, the over-provisioning of resources on [Amazon Elastic Compute Cloud] (Amazon EC2) can lead to unnecessary costs and resource consumption.

Here are a few recommendations to help you optimize costs and lower the TCO for your Windows Server workloads on Amazon EC2:

- **Identify the right instance type.** Do the free AWS Optimization and Licensing Assessment to evaluate your Windows environment and identify ways to reduce licensing costs and run your resources more efficiently.
- **Use optimization tools for Windows Server workloads.** AWS offers a broad range of tools that can help you gain insight into your current and future AWS costs. We recommend AWS Compute Optimizer, AWS Trusted Advisor, and AWS Cost Explorer.
- **Select the right Savings Plans.** Savings Plans are a flexible pricing model that can help you reduce your bill by up to 72 percent compared to On-Demand prices in exchange for a one- or three-year hourly spend commitment.

Read for more cost-saving recommendations for Windows Server >

Can we automate the migration of our Microsoft workloads to AWS?

Yes, we have automation tools to help with your migration.

AWS Application Migration Service
This is a free recommended service to help you migrate to AWS. It simplifies and expedites your migration by automatically converting your source servers from physical, virtual, or cloud infrastructure to run natively on AWS. It further simplifies your migration and reduces costs by allowing you to use the same automated process for a wide range of applications without changes to the applications, their architecture, or the migrated servers.

AWS Migration Hub Strategy Recommendations
The AWS Migration Hub helps you easily build a migration and modernization strategy for your applications running on premises or in the AWS Cloud. The Strategy Recommendations feature automates the manual process of analyzing each running application—its process dependencies and technical complexity. This helps reduce the time and effort spent on planning and speeds up your business transformation on AWS.

AWS Migration Hub Orchestrator
The Orchestrator feature of the AWS Migration Hub provides a single location to run and track your migrations. The Orchestrator simplifies and automates the migration of servers and enterprise applications to AWS and provides templates to create customizable migration workflows to fit your unique migration requirements.

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How can we optimize and modernize our SQL Server workloads on AWS?

We know that the high cost of Microsoft licenses for SQL Server, especially SQL Server Enterprise edition, is a point of concern for many of our customers.

Here are a few money-saving recommendations for SQL Server on Amazon EC2 deployments:

- Consolidate small SQL Server databases
- Use SQL Server Developer edition for non-production environments
- Use the Optimize CPU feature to save up to 75 percent

You can also deploy SQL Server on Amazon Relational Database Service (Amazon RDS). You get a fully managed relational database service that makes it easy for you to set up, operate, and scale your SQL Server deployment while significantly reducing your operational overhead.

In addition to supporting SQL Server, AWS offers the widest variety of databases that are purpose-built for different types of applications so that you can choose the right tool for the job to get the best cost and performance. We also offer open-source alternatives, such as SQL Server on Linux and Babelfish for Aurora PostgreSQL.

Read for more cost-saving recommendations for SQL Server >

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Can we run our .NET applications on AWS?

Yes, you can; in fact, we have been supporting .NET in the cloud since 2008. You can run both your legacy .NET Framework and modern .NET applications on AWS.

Here are some of the ways you can run .NET on AWS:

Migrate
The simplest way to migrate .NET applications to AWS is to rehost them using **AWS Elastic Beanstalk** or **Amazon Elastic Cloud Compute** (Amazon EC2).

Containers
With containers, you can bundle your .NET application with its dependencies and configuration, making it easy to port between on premises and the cloud. AWS offers a variety of container services to host your containerized .NET Framework application on Windows or your modern .NET application on Linux. You can use the **AWS App2Container** tool to generate a container image for your application.

Modernize
You can modernize your .NET application to a cloud-based architecture to maximize scale and reliability, take advantage of serverless compute, and run on Linux at a reduced cost. The **Porting Assistant for .NET** tool helps you port your code to modern .NET, and the **AWS Microservice Extractor** for .NET tool simplifies the process of transforming your monolithic application into microservices.

For additional guidance, tools, and community connections, visit:
.NET on AWS developer hub >

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Can we run Windows Server containers on AWS?

Yes, you can run Windows Server containers on AWS, and here's why you should.

1. Containers are more efficient than virtual machines. They provide better isolation and let you maximize the placement of your applications to increase the utilization of your infrastructure resources and reduce overall costs.

2. It's easier to automate the stages of the development lifecycle (build, test, and deploy). By automating these processes, DevOps teams have more time to focus on innovating.

3. You can modernize legacy applications by using Windows Server containers and running them on a modern infrastructure. This will help you improve the scalability, security, and maintainability of your legacy applications.

4. Containers allow your applications to scale up rapidly, making them useful for applications that require fast expansion or cloud bursting.

AWS offers free immersion days to help upskill your development teams on container technology and DevOps practices.

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How can AWS help us handle end-of-support challenges for Windows Server and SQL Server?

When it comes to migrating, optimizing, and modernizing end-of-support (EOS) workloads, it is imperative to evaluate cloud providers holistically.

Forced decision points like EOS events create a unique opportunity for you to think about the future state of your business, evaluate options and alternatives, and determine which path is right for you. When evaluating cloud providers, you will want to consider such factors as price performance, reliability, breadth and depth of technology, pace of innovation, and flexible licensing options.

Here are a few of the options you have with AWS:

- **Assess, optimize, migrate, and take advantage of automated upgrades.**
  Start with the free AWS Optimization and Licensing Assessment to assess and optimize your current on-premises and cloud environments. After your optimized migration, automate your upgrades with AWS Systems Manager automation runbooks for our EOS workloads.

- **Future-proof your legacy Windows Server applications.**
  To address compatibility and dependency issues, AWS offers the End-of-Support Migration Program for Windows Server to help you migrate your legacy applications to the latest supported versions of Windows Server on AWS without any code changes.

- **Modernize your workloads to accelerate innovation with open-source and cloud-based technologies.**
  Accelerate the modernization of your legacy infrastructure, databases, and applications to open-source and cloud-based technologies, and break free from the software upgrade and refresh cycle.

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What is your top recommendation for running Microsoft workloads on AWS?

Our top recommendation is to optimize your Microsoft workloads—from both a licensing and infrastructure standpoint—to run in the cloud. If you simply lift and shift your workloads from on premises to the cloud without optimizing, you may miss out on the savings and efficiencies the cloud provides.

For license optimization, AWS offers a free **Optimization and Licensing Assessment** (AWS OLA) program for both new and existing customers. The AWS OLA gives you the opportunity to assess and optimize your on-premises and cloud environments based on actual resources used and third-party licensing.

By understanding your current hardware, application performance, and existing contracts, AWS can provide you with a migration and licensing strategy that will rightsize your resources, provide a clear AWS road map, and eliminate costs by using your existing investments and making sure you're only paying for what you need.

If your Microsoft Enterprise Agreement is coming up for renewal or you're running Windows Server or SQL Server versions that are reaching (or have reached) the end of support, now is a great time to do an AWS OLA.

Ongoing infrastructure optimization is also key to achieving continuous savings. Our customers use the **AWS Compute Optimizer** to make sure they are provisioning instances to match their exact workload demands. This service analyzes the configuration and utilization metrics of your AWS resources, reports whether your resources are optimal, and generates optimization recommendations to reduce the cost and improve the performance of your workloads.

Transform your business. Use AWS to migrate your Microsoft workloads.

AWS is the world's most comprehensive and broadly adopted cloud platform, offering over 200 fully featured services from data centers globally. Millions of customers—including the fastest-growing startups, largest enterprises, and leading government agencies—are using AWS to lower costs, become more agile, and innovate faster.

AWS has significantly more services and more features within those services than any other cloud provider, making it faster, easier, and more cost-effective to move your existing applications to the cloud and build nearly anything you can imagine. Whether you need to deploy your application workloads around the globe in a single click or you want to build and deploy specific applications closer to your end users with single-digit millisecond latency, AWS provides you with cloud infrastructure where and when you need it.

Give your Microsoft applications the infrastructure they need to drive the business outcomes you want.

Let’s get started.

Get more details and deepen your understanding of running Microsoft workloads on AWS by visiting aws.amazon.com/windows.